

Kendall's Tau

2026-04-17

Introduction

Kendall's τ quantifies association via the proportion of pairs of observations that are **concordant** (both variables increase or both decrease) minus the proportion that are **discordant**. It is more robust than Spearman in small samples and under ties.

Prerequisites

Ranks, concordant / discordant pairs.

Theory

For n observations, there are $\binom{n}{2}$ pairs. A pair (i, j) is concordant if $\text{sign}(X_i - X_j) = \text{sign}(Y_i - Y_j)$, discordant if the signs differ, and tied otherwise.

$$\tau = \frac{\text{concordant} - \text{discordant}}{\binom{n}{2}}.$$

Variants (for ties):

- **tau-a**: simple difference divided by all pairs (including ties).
- **tau-b**: corrects for tied pairs in both variables; most commonly used with tied data.
- **tau-c** (Stuart's tau-c): for different-sized marginal distributions.

$\tau \in [-1, 1]$; 0 indicates no association. The null distribution (no association) is asymptotically normal; exact distributions exist for small n .

Assumptions

Same as Spearman: ordinal or continuous data, monotonic relationship, independent pairs.

R Implementation

```
library(DescTools)
set.seed(2026)

n <- 40
x <- rnorm(n)
y <- 0.6 * x + rnorm(n, sd = sqrt(1 - 0.6^2))

cor.test(x, y, method = "kendall")

# Spearman for comparison
cor.test(x, y, method = "spearman")

# Tau-b (explicit)
```

```
KendallTauB(x, y, conf.level = 0.95)

# With ties
x2 <- round(x, 1)
y2 <- round(y, 1)
cor.test(x2, y2, method = "kendall")
```

Output & Results

Kendall's rank correlation tau

```
data: x and y
z = 3.68, p-value = 0.0002
alternative hypothesis: true tau is not equal to 0
tau: 0.394
```

Spearman's rho: 0.57

Both correlations are significantly positive; Kendall's τ is numerically smaller than Spearman (typical, as they measure different things).

Interpretation

Report: “Kendall's $\tau = 0.39$ ($z = 3.68$, $p < 0.001$), indicating a moderate positive monotonic association between x and y .”

Practical Tips

- Kendall and Spearman measure different quantities; do not treat them as interchangeable.
- Kendall is more robust in small samples; Spearman is more common in reporting.
- For heavily tied data (e.g., Likert scales), prefer Kendall tau-b over Spearman.
- `DescTools::KendallTauB()` provides the tau-b with a CI, which base R's `cor.test` does not.
- Kendall tau has a direct interpretation as (probability of concordance - probability of discordance), which Spearman does not.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation

- Non-parametric tests